Small speech models have quietly become the most useful part of the local stack. A forty million parameter network, quantised to eight bits, occupies less space than a single photograph, yet it can render a paragraph of text as natural speech in a fraction of the time a cloud service needs to accept the request. This is not a marginal improvement. It changes what an application can assume about latency, privacy, and cost, and it therefore changes which applications are worth building at all.

Consider the reading machine. Dr. Smith, who runs a clinic in a small town, does not need a general purpose assistant. She needs a device that reads the small print on a pill bottle out loud, in the patient's own language, without sending a photograph of a prescription to a data centre somewhere else. The constraint is not intelligence. The constraint is that the round trip must finish before the patient loses patience, e.g. within about a second, and that no image ever leaves the room.

The same argument applies to long form reading. A person who wants to hear a two hundred page report while walking the dog is not asking for a summary. They are asking for the whole thing, spoken at a steady pace, with the paragraph breaks preserved and the footnotes skipped. Doing this well is mostly a text processing problem. A PDF is a container for positioned glyphs, not for sentences, so the pipeline must reconstruct the sentences first: remove the running head, drop the page numbers, undo the hyphenation that the typesetter inserted to justify a line, then cut the result into chunks that a speech model can handle without losing the thread of the argument.

There is a second reason to keep the model small. Synthesis at a hundred times real time means a one hour audiobook renders in under a minute, which makes batch generation practical on a laptop. A publisher can produce six voice variants of the same chapter, listen to all of them, and keep the one that sounds least like a machine reading a telephone directory.

None of this is free of caveats. The licensing matters: the multilingual weights carry a non commercial restriction, so a paid product must either use the permissively licensed base model or negotiate a separate agreement. The language coverage is narrow. Five European languages are supported, and a document in any other script will produce confident nonsense rather than an error, which is the worst possible failure mode for a reading aid.

The voice catalog is fixed. There is no cloning, no emotion tag, no way to make the narrator sound like a particular person. For an audiobook that is a real limitation; for a notification system it is irrelevant. The lesson is that a small model should be placed where its constraints do not matter, and that is usually in the middle of a pipeline rather than at the end of one.